

LECTURE 15: MATH1061/7861
TOPICS: STRONG MATHEMATICAL INDUCTION AND
THE WELL-ORDERING PRINCIPLE
DATE: THURSDAY MARCH 26, 2026

Learning objectives:

- (1) Apply the principles of mathematical induction and the strong version.
- (2) Reinforce how the basis step relates to the inductive step in mathematical induction and in the strong version.
- (3) Understand the well ordering principle to prove something.

Q1:

Let $a_0, a_1, a_2, \dots$ be a sequence defined by

$$a_0 = 6, \quad a_1 = 9, \quad a_n = a_{n-1} + a_{n-2} \quad \forall n \geq 2.$$

Prove that $3 \mid a_n$ for all $n \geq 0$.

$$a_2 = a_1 + a_0 = 9 + 6 = 15$$

$$a_3 = a_2 + a_1 = 15 + 9 = 24$$

$$a_4 = a_3 + a_2 = 24 + 15 = 39$$

All divisible
by 3.

Base Cases: $a_0 = 6$ and $a_1 = 9$ are both divisible
by 3.

Inductive hypothesis: Suppose statement is true for
 $n = 0, 1, \dots, K$. I.e. $3 \mid a_0, 3 \mid a_1, \dots, 3 \mid a_{K-1}, 3 \mid a_K$

what we want to prove: $3 \mid a_{K+1}$

Inductive step: By definition $a_{K+1} = a_K + a_{K-1}$

By inductive hypothesis $3 \mid a_K$ and $3 \mid a_{K-1}$.

This means $a_K = 3x$ and $a_{K-1} = 3y$

for some $x, y \in \mathbb{Z}$. So

$$a_{K+1} = a_K + a_{K-1} = 3x + 3y = 3(x+y)$$

so $3 \mid a_{K+1}$.

Q2:

Let $b_1, b_2, b_3, \dots$ be defined by

$$b_1 = 2, \quad b_2 = 7, \quad b_n = 2b_{n-1} + 3b_{n-2} \quad \forall n \geq 3.$$

Prove that $b_n < 5^n$ for all $n \geq 1$.

✓ Base Cases: $b_1 = 2 < 5^1$ $b_2 = 7 < 25 = 5^2$

Inductive hypothesis Suppose $b_n < 5^n$ for all $n = 1, 2, \dots, K$. That is $b_1 < 5^1, \dots, b_{K-1} < 5^{K-1}, b_K < 5^K$.
What we want to prove $b_{K+1} < 5^{K+1}$

Inductive Step: By definition $b_{K+1} = 2b_K + 3b_{K-1}$

By inductive hypothesis $b_K < 5^K$ and $b_{K-1} < 5^{K-1}$.

Thus, $b_{K+1} = 2b_K + 3b_{K-1}$

(By inductive hypo) $< 2 \times 5^K + 3 \times \underline{5^{K-1}}$

$(5^{K-1} < 5^K)$ $\ll 2 \times 5^K + \underline{3 \times 5^K}$
 $= 5 \times 5^K = \underline{5^{K+1}}$

Not Assessable

Every integer bigger than 1 can be written as a product of primes.

Proof: Suppose, for the sake of contradiction, there exists an integer bigger than 1 which cannot be written as a product of primes. By the well-ordering principle, we can find the smallest such an integer. Let us call it n .

If n is a prime, then we have a contradiction (because a prime is a product of primes).

If n is not a prime, then n is composite.

So we can write $n = ab$ $a, b \in \mathbb{Z}_{>1}$.

Now $1 < a < n$ and $1 < b < n$.

By minimality of n , a and b are both products of primes. But then $n = ab$ is also a product of primes. This is a contradiction.